Real-Time Operating System (RTOS)

Learning Outcomes

- Defining a Real-Time Operating System.
- Concept of deadlines and response time.
- Differentiate **Hard, Firm, and Soft Real-Time Systems**.
- Important characteristics of RTOS.
- Identifying real-world RTOS applications.
- Understanding the difference between a general-purpose OS and RTOS.

What is a Real-Time System?

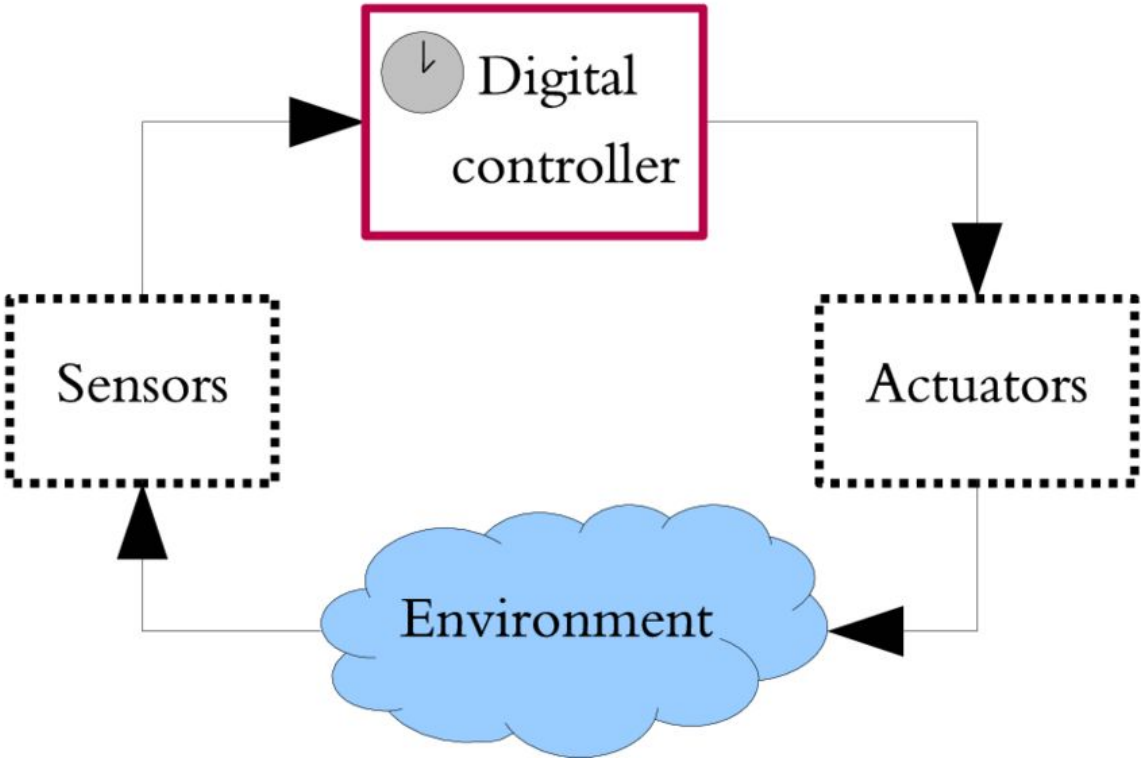
Definition
A **Real-Time System** is a computer system in which the correctness of an operation depends on:

1. The correctness of the result
2. The time at which the result is produced

Key Concept
Correct Result + Correct Time = Correct Operation

Example:
Airbag system:
Sensor detects collision → System processes information → Airbag deploys
The response must occur within a strict time limit.

In a normal system, we usually care mainly about getting the correct result.
In a real-time system, we care about both the result and **when that result is produced**.


Digital
controller
Sensors
Actuators
Environment

What is RTOS?

Real-Time Operating System

An RTOS is an operating system designed to provide **predictable** **and** **timely responses to events**.

Important Characteristics

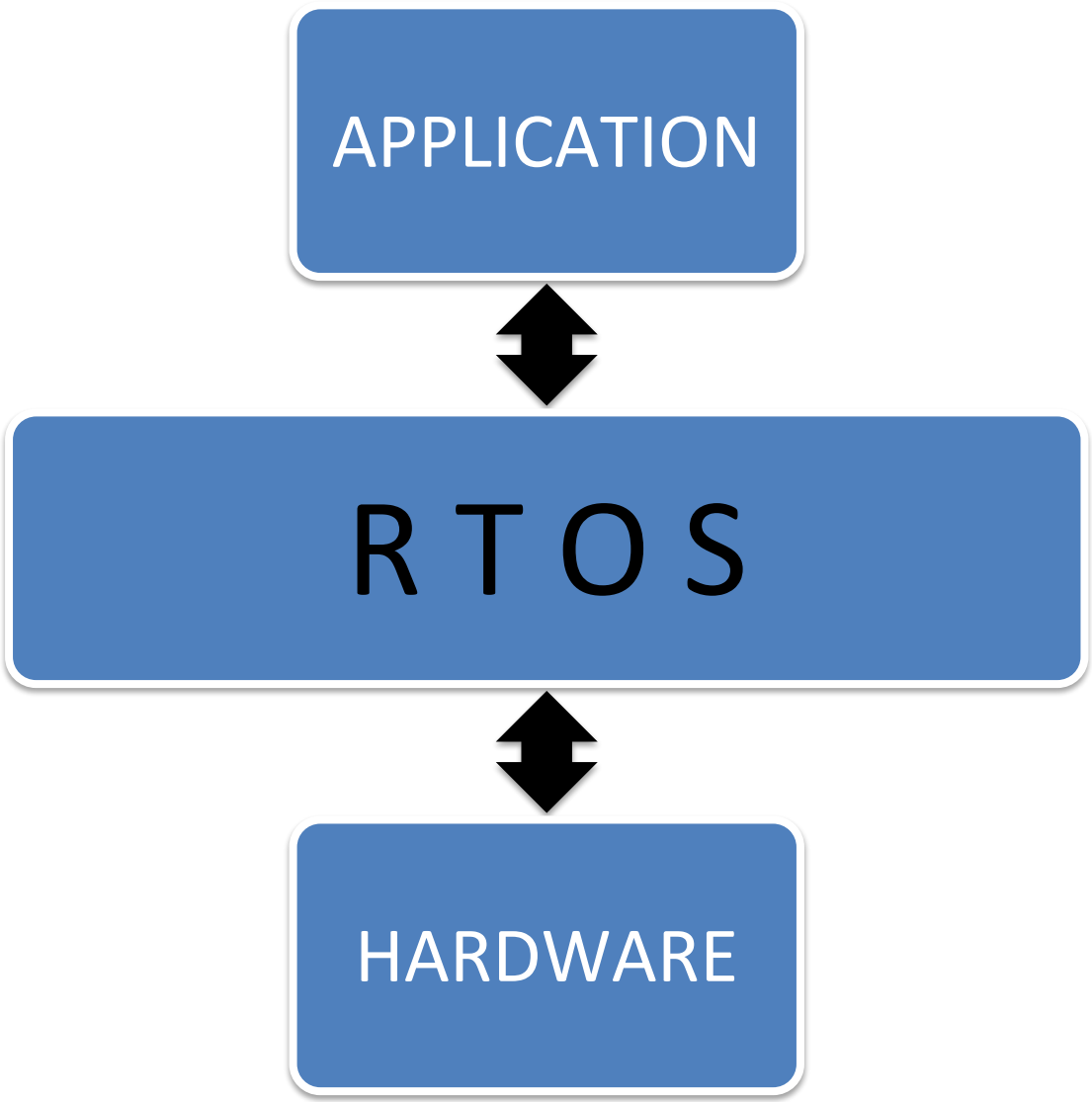
- Predictable response time
- Deterministic behavior
- Priority-based scheduling
- Low interrupt latency
- Fast context switching
- Reliability

Examples

- FreeRTOS
- VxWorks
- QNX
- Zephyr

Keywords:

1. Immediate
2. Quick
3. Deadline


APPLICATION
R T O S
HARDWARE

Why Do We Need RTOS?

Consider These Systems

- ✈ Aircraft control
- 🚗 Vehicle braking
- 🏥 Medical equipment
- 🤖 Industrial robots
- 🚀 Spacecraft
- 🏭 Industrial automation

Question

What happens if the response is late?

In these systems, a delayed response can cause equipment failure, financial loss, or even danger to human life.
Therefore, we need systems where the timing requirements can be guaranteed.

General Purpose OS vs RTOS

FEATURE	GENERAL-PURPOSE OS	RTOS
Main goal	Convenience & throughput	Predictable timing
Response	Variable	Predictable
Scheduling	General-purpose	Often priority-based
Deadline	Usually not strict	Important
Example	Windows, macOS	FreeRTOS, VxWorks

Windows and macOS are excellent operating systems, but they are not designed to guarantee that a particular task will always finish before a strict deadline.
RTOS is designed around these timing requirements.

Important RTOS Terms

1. Task
A unit of work executed by the system.
2. Deadline
The maximum allowable time by which a task must complete.
3. Response Time
Time between an event occurring and the system responding.
4. Determinism
Ability to respond within a predictable time.

Example
Sensor event occurs at:
10:00:00.000
Required response before:
10:00:00.010
Deadline = **10 ms**

Types of Real-Time Systems

Three Important Categories

1. Hard Real-Time:

Deadline **must** be met.

2. Firm Real-Time:

Late result has **no value**, but occasional deadline misses may be tolerated.

3. Soft Real-Time:

Deadline is important, but occasional delay is acceptable.

Hard Real-Time System

Definition

A system in which **missing a deadline can cause system failure or unacceptable consequences.**

Examples

- Aircraft control systems
- Airbag systems
- Pacemakers
- Nuclear reactor control

Characteristics

- Strict deadlines
- Predictable response
- High reliability
- Deadline violations are unacceptable

Imagine an airbag system:

If the system detects a crash but deploys the airbag several seconds later, the system has failed. Therefore, this is a hard real-time requirement.

Soft Real-Time System

Definition
A system where missing a deadline occasionally **reduces performance or quality**, but does not necessarily cause system failure.

Examples
- Video streaming
- Online gaming
- Multimedia
- Voice communication

Example
If a video frame arrives slightly late:

Result: Small interruption or reduced quality.
The entire system does not fail.

Firm Real-Time System

Definition
A firm real-time system has deadlines where a result produced after the deadline is **useless**.
However, occasional deadline misses may be tolerated.

Example
A real-time image-processing system:
Image arrives → Processing → Result
If processing finishes after the required deadline, that particular result may no longer be useful.

Hard vs Firm vs Soft

FEATURE	HARD	FIRM	SOFT
Deadline	Strict	Important	Important
Late result	Unacceptable	Useless	Still useful
Deadline miss	System failure	Result discarded	Performance decreases
Example	Airbag	Industrial inspection	Video streaming

Important Exam Question
Differentiate Hard, Firm and Soft Real-Time Systems.

RTOS Scheduling

Scheduling
The OS decides:

Which task should execute next?
RTOS commonly uses **priority-based scheduling**.

Task	Priority
Emergency Sensor	1 – Highest
Motor Control	2
Data Logging	3
Display Update	4

Higher-priority tasks receive CPU attention first.

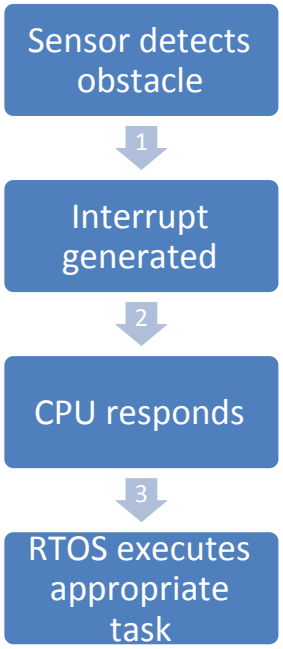
Imagine a robot:
If the emergency stop sensor generates an event, that task should be handled before updating the display.
Therefore, priority becomes extremely important.

Interrupts in RTOS

Interrupt

An interrupt is a signal that informs the CPU that an event requires attention.

Example:



Key Term

Interrupt Latency

Time between the occurrence of an interrupt and the beginning of its handling.

RTOS Goal:

Low and predictable interrupt latency

Context Switching

Context Switch
The process of saving the state of one task and loading the state of another task.

Example
Task A
↓
Context Switch
↓
Task B

RTOS Requirement
- Fast context switching
- Predictable execution time

Suppose Task A is running and a higher-priority Task B becomes ready.
The RTOS may pause Task A, save its state, and start Task B.
This switching is called a context switch.

Characteristics of RTOS

Major Characteristics:

- Deterministic behavior
- Predictable response time
- Priority-based scheduling
- Low interrupt latency
- Fast context switching
- Reliability
- Efficient resource utilization
- Multitasking

Important Point:
RTOS ≠ Simply Fast OS

Remember
Predictability is more important than raw speed.

Applications of RTOS

Real-World Applications

Automotive

- Airbags
- ABS
- Engine control

Medical

- Patient monitoring
- Medical instruments

Aerospace

- Flight control
- Navigation

Industrial

- Robots
- Manufacturing systems

Consumer Electronics

- Cameras
- Smart appliances

RTOS Examples

Popular RTOS:

FreeRTOS
- Widely used in embedded systems
- Lightweight
- Open-source

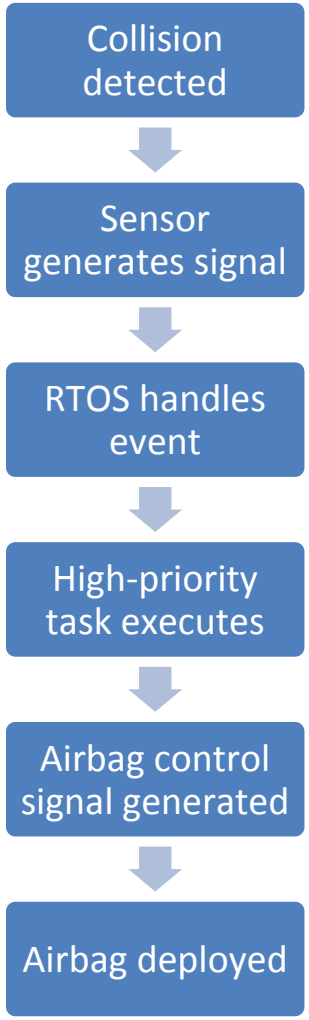
VxWorks
- Used in embedded and mission-critical systems

QNX
- Used in automotive and embedded systems

Zephyr
- Open-source RTOS
- Designed for resource-constrained devices

Case Study

🚗 **Car Airbag System Sequence**



What type of real-time system is this?

Q1. What is the most important characteristic of an RTOS?

A. Attractive user interface. B. Predictable response time.
C. Large storage capacity. D. High graphics performance

Q2. Which system has the strictest deadline requirement?

A. Soft Real-Time B. Firm Real-Time. C. Hard Real-Time. D. Batch System

Q3. Which is an example of a Hard Real-Time System?

A. Video streaming. B. Online gaming. C. Airbag control. D. Web browsing

Q4. What is the time between an interrupt occurring and its handling called?

A. Execution time. B. Interrupt latency. C. Waiting time. D. Turnaround time

Q5. Which statement is correct about an RTOS?

A. It is always faster than every general-purpose OS. B. It only supports one task
C. It provides predictable and timely responses. D. It cannot use multitasking

B
C
C
B
C